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Continuous Optimization Using Evolutionary Strategies

1. Introduction

In this project, an Evolutionary Strategy (ES) is applied to solve a continuous global optimization task based on the two-dimensional Rastrigin function. The Rastrigin function is a classic benchmark for optimization algorithms, it features a large number of local minima arranged in a grid-like pattern around a single global minimum at the origin, making it an ideal stress-test for methods that can easily become trapped by local search behavior [1, 2]. The goal of this project is not only to obtain a solution at or near the global optimum within the bounded domain $(-5.12, 5.12)^2$, but also to evaluate how key ES design choices; particularly the parent-to-offspring ratio (μ:λ), individual versus shared step sizes (σ), and the use of recombination operators; affect solution quality, stability across different random seeds, and computational cost [3]. By conducting a structured set of 35 runs with progressively more sophisticated ES variants, this assigment provides insight into the exploration-exploitation tradeoff inherent in evolutionary strategies and quantifies which design choices yield the most reliable convergence toward the global minimum.

2. Methodology, Definitions and Assumptions

2.1. Problem definition

The optimization target is the two-dimensional Rastrigin function, defined as:

$$f(x_1, x_2) = 20 + (x_1^2 - 10\cos(2\pi x_1)) + (x_2^2 - 10\cos(2\pi x_2))$$

Each variable is bounded to the range [-5.12, 5.12]. The function has a global minimum of f(0, 0) = 0.0 at the origin. The highly multimodal landscape, with approximately 50 local minima within the search domain, which makes this problem challenging for optimization.

2.2. Solution encoding

Each individual in the ES population is encoded as a real-valued vector $x = (x_1, x_2)$ representing a candidate solution. Also, each individual carries one or more strategy parameters (σ) that control the mutation step size. Two encoding schemes are used across experiments: (1) <u>Shared σ</u>: A single standard deviation σ is used for both variables; (2) <u>Individual σ</u>: Each variable has its own step size σ_1, σ_2, allowing direction-dependent adaptation.

2.3. (μ, λ) ES selection

A comma strategy (μ, λ)-ES is adopted, where λ offspring are generated from μ parents each generation, and the best μ offspring are selected as the next generation's parents. The comma strategy is preferred over the plus strategy for continuous optimization because it allows the population to forget previously good but potentially misleading solutions, which improves exploration on multimodal landscapes like the Rastrigin function.

2.4. Mutation and the 1/5 success rule

Offspring are generated by adding Gaussian noise to parent solutions: $x' = x + \sigma \cdot N(0, 1)$. The step size σ is adapted using Rechenberg's 1/5 success rule: every 10 generations, the fraction of successful mutations (offspring better than the median parent cost) is computed. If the success rate exceeds 1/5, σ is increased by dividing by a damping factor c = 0.82 (encouraging larger steps); if it falls below 1/5, σ is decreased by multiplying by c; if exactly 1/5, σ remains unchanged. This self-adaptive mechanism allows the ES to balance exploration (large σ) and exploitation (small σ) throughout the search.

2.5. Recombination operators

Two recombination operators are investigated in the later experiments: (1) Global Intermediate Recombination (on σ): For each offspring, two random individuals are selected and their σ values are averaged to produce the offspring's σ. This smooths step-size adaptation across the population. (2) Dual Discrete Recombination (on x): For each variable in the offspring, the value is randomly copied from one of two randomly selected individuals. This introduces diversity by mixing decision variable components.

2.6. Population initialization

The initial parent population consists of μ individuals drawn uniformly at random from the search domain $[-5.12, 5.12]^2$. The initial step size is set to $\sigma_0 = 1$. Five independent random seeds (1, 2, 3, 4, 5) are used to evaluate sensitivity to starting conditions and ensure reproducibility.

2.7. Stopping criterion

All runs use a fixed stopping criterion of 500 generations. This provides sufficient time for the ES to converge while keeping computational cost manageable.

2.8. Experimental design

To evaluate the progressive improvement of ES design choices, five experiments are conducted. The detail is mentioned in the Table 1. Experiment B tests three μ:λ ratios: (5, 35), (15, 100), and (30, 200), each run with 5 seeds = 15 runs. The best ratio from Experiment B is then used for all subsequent experiments (C, D, E).

Table 1 Detail of the experiments (35 total runs across 5 ES variants)

Experiment	σ Encoding	σ Adaptation	Recombination	Runs
A: Basic ES	Single σ	1/5 rule	None	5
B: μ:λ sweep	Single σ	1/5 rule	None	15
C: Individual σ	Per-variable σ	1/5 rule (each)	None	5
D: σ Recomb	Per-variable σ	1/5 rule + recomb	Global interned	5
E: x Recomb	Per-variable σ	1/5 rule	Dual discrete on x	5

3. Results and Discussion

3.1. Rastrigin function visualization

Figure 1 shows the 3D surface and contour plot of the 2D Rastrigin function. The function's highly multimodal nature is clearly visible, with numerous regularly spaced local minima surrounding the global minimum at the origin.

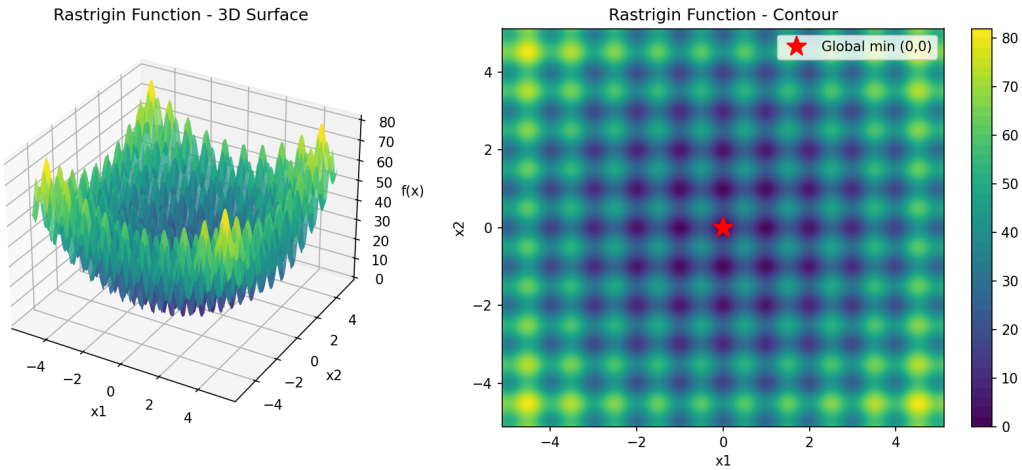

Figure 1 Rastrigin function, 3D surface and contour plot

3.2. Single run performance

The initial single run of the (μ, λ) for ES was executed with μ=15 parents, λ=100 offspring, a single shared σ (initial $\sigma_0 = 1.0$), the 1/5 success rule, and no recombination over 500 generations using seed 42. The ES produced a best cost of approximately 0.0 ($f \approx 7.9\times10^{-10}$) at $x^* \approx (1.4\times10^{-6}, -1.4\times10^{-6})$, effectively reaching the global optimum. Figure 2 and Figure 3 shows the results for the single run.

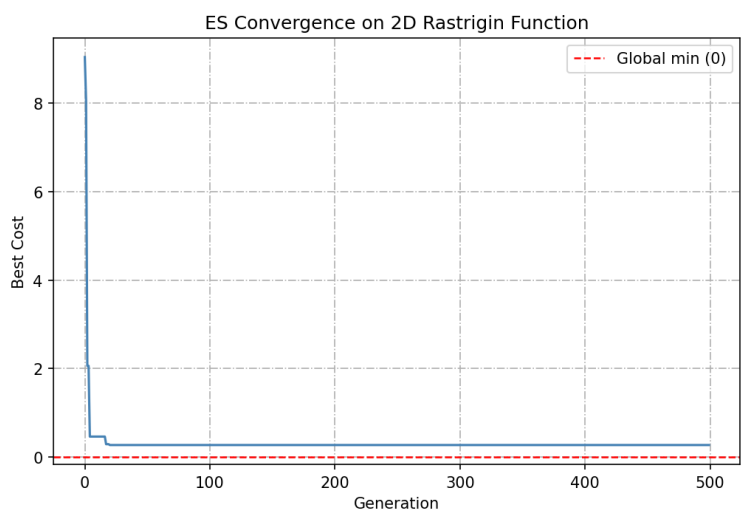

Figure 2 Single run convergence curve over 500 generations

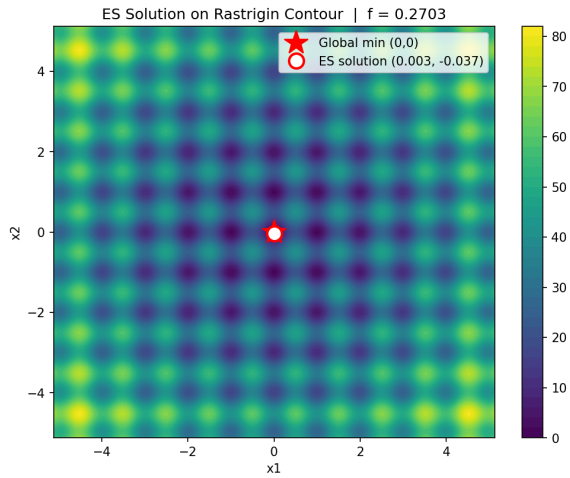

Figure 3 ES solution on the Rastrigin contour plot

3.3. Experiment A: basic ES

The basic ES was run with 5 different seeds using μ=15, λ=100. Four of the five seeds found solutions extremely close to the global optimum ($f < 10^{-9}$), while seed 3 converged to a local minimum with $f \approx 0.2269$. The mean cost across all 5 runs was 0.0454 with a standard deviation of 0.1015. The results are in the Table 2.

Table 2 Experiment A results, Basic ES with 5 seeds

Seed	Best x_1	Best x_2	Best Cost	Runtime (s)
1	1.40E-06	-1.43E-06	7.91E-10	0.149
2	1.82E-06	1.07E-06	8.83E-10	0.136
3	2.58E-02	-2.19E-02	2.27E-01	0.135
4	-5.37E-07	7.33E-07	1.64E-10	0.136
5	-3.99E-08	2.01E-06	8.04E-10	0.133

3.4. Experiment B: μ:λ ratio sweep

Three μ:λ ratios were tested: (5, 35), (15, 100), and (30, 200). The ratio 30:200 consistently outperformed the other two configurations, achieving near-zero cost on all 5 seeds. The 5:35 ratio showed the highest mean cost and greatest variance. Table 3 and Figure 4 shows the detail of each seed. Figure 4 uses a logarithmic scale to highlight the orders of magnitude difference between ratios; which the 30:200 ratio achieved costs around 10^{-10}, while the 5:35 and 15:100 ratios had outliers near 10^{-1} (local minima traps).

Table 3 Mean cost by μ:λ ratio across 5 seeds

μ:λ Ratio	Mean Cost	Std	Min Cost	Max Cost
5:35	6.24E-02	8.77E-02	2.20E-09	2.12E-01
15:100	4.54E-02	1.01E-01	1.64E-10	2.27E-01
30:200	1.50E-10	9.48E-11	5.92E-11	2.84E-10

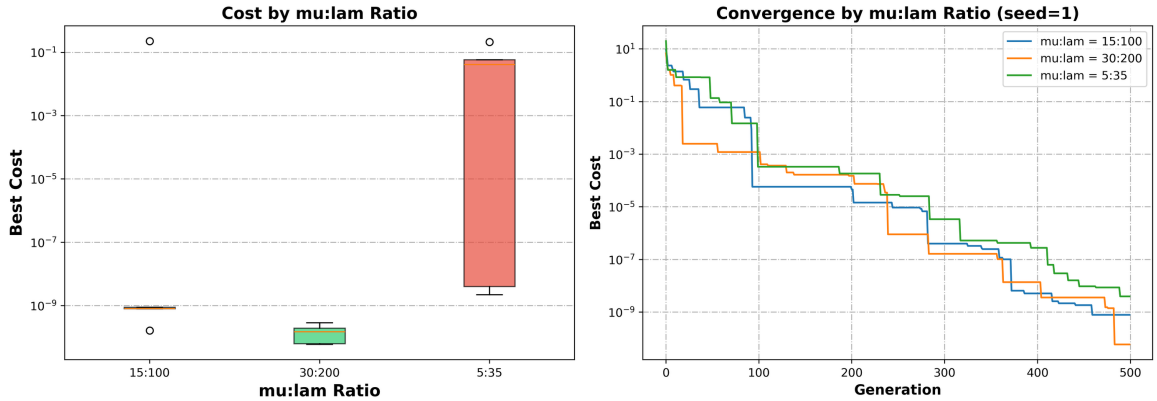

Figure 4 μ:λ ratio analysis — boxplot and convergence comparison

Based on these results, the best μ:λ ratio of 30:200 was selected for all subsequent experiments. The full results of the 15 runs are shown in the Table 4.

Table 4 Results of the experiment B (15 runs)

Run id	seed	μ	λ	μ:λ ratio	Best x_1	Best x_2	Best cost	Runtime (s)
6	1	5	35	5:35	4.43E-06	5.89E-07	3.96E-09	0.053
7	2	5	35	5:35	0.012653	0.011538	0.058149	0.055
8	3	5	35	5:35	-0.0091	-0.03149	0.212475	0.055
9	4	5	35	5:35	3.09E-06	1.25E-06	2.2E-09	0.053
10	5	5	35	5:35	-0.0143	-0.00203	0.041357	0.052
11	1	15	100	15:100	1.4E-06	-1.4E-06	7.91E-10	0.137
12	2	15	100	15:100	1.82E-06	1.07E-06	8.83E-10	0.137
13	3	15	100	15:100	0.025795	-0.02192	0.226909	0.140
14	4	15	100	15:100	-5.4E-07	7.33E-07	1.64E-10	0.133
15	5	15	100	15:100	-4E-08	2.01E-06	8.04E-10	0.131
16	1	30	200	30:200	1.4E-07	5.28E-07	5.92E-11	0.250
17	2	30	200	30:200	-5E-07	-2.5E-07	6.21E-11	0.258
18	3	30	200	30:200	-1.1E-06	5.26E-07	2.84E-10	0.252
19	4	30	200	30:200	-5.1E-08	-9.9E-07	1.94E-10	0.251
20	5	30	200	30:200	8.66E-07	4.64E-08	1.49E-10	0.252

3.5. Experiment C: Individual σ per variable

Using the best ratio (30:200), individual step sizes σ_1 and σ_2 were assigned to each variable, each adapted independently via the 1/5 success rule. All 5 seeds converged to the global optimum with mean cost 1.4975e-10 and standard deviation 9.4752e-11 (Table 5). For a two-variable problem, the benefit of individual σ is limited since the landscape is symmetric in x_1 and x_2, but this variant provides the foundation for the recombination experiments that follow.

Table 5 Results of the experiment C (5 runs)

Run id	seed	Best x_1	Best x_2	Best cost	Runtime (s)
21	1	1.40206E-07	5.28067E-07	5.92237E-11	0.240
22	2	-5.01135E-07	-2.48807E-07	6.2105E-11	0.237
23	3	-1.0749E-06	5.26389E-07	2.84196E-10	0.239
24	4	-5.08489E-08	-9.87593E-07	1.94014E-10	0.238
25	5	8.66055E-07	4.63572E-08	1.49232E-10	0.240

3.6. Experiment D: Global intermediate recombination on σ

Building on Experiment C, global intermediate recombination was applied to the step sizes: for each offspring, σ is set to the average of two randomly selected parents' σ values. The 1/5 rule is applied first, and then recombination is performed. All 5 seeds converged to the global optimum with mean cost 2.4462e-10 (Table 6). The best run across all 35 experiments came from this variant (seed 5, f* = 1.8122e-11). The mean runtime increased to 0.668s due to the additional recombination computation.

Table 6 Results of the experiment D (5 runs)

Run id	seed	Best x_1	Best x_2	Best cost	Runtime (s)
26	1	-8.8E-07	-1.4E-06	5.25E-10	0.667
27	2	5.14E-07	3.31E-07	7.43E-11	0.665
28	3	-8.1E-07	7.15E-07	2.32E-10	0.650
29	4	-6.8E-07	-1.2E-06	3.74E-10	0.673
30	5	-2.5E-08	3.01E-07	1.81E-11	0.724

3.7. Experiment E: Dual discrete recombination on x

In this final variant, dual discrete recombination is applied to the decision variables x_1 and x_2. Individual σ from Experiment C is retained (without σ recombination). The next generation is created first using the standard mutation approach, and then recombination is applied. All 5 seeds converged to the global optimum with mean cost 3.8439e-10 and standard deviation 3.7113e-10 (Table 7). The mean runtime was 0.833s.

Table 7 Results of the experiment E (5 runs)

Run id	Best x_1	Best x_2	Best x_1	Best x_2
31	8.31395E-07	7.91409E-07	2.61E-10	0.986
32	3.66097E-07	-1.81644E-07	3.31E-11	0.829
33	-5.75402E-07	-1.1176E-07	6.82E-11	0.835
34	1.94662E-06	3.57714E-07	7.77E-10	0.815
35	-1.82304E-06	-7.86558E-07	7.82E-10	0.828

3.8. Comparison across all experiments

Figure 5 presents boxplots comparing the best cost achieved by each ES variant. Experiments A and B show occasional outliers where the ES became trapped in local minima, while Experiments C, D, and E consistently achieved costs below 10^{-9}. The logarithmic scale in shows that Experiments C, D, and E achieved costs approximately 8 orders of magnitude lower than the outliers in A and B.

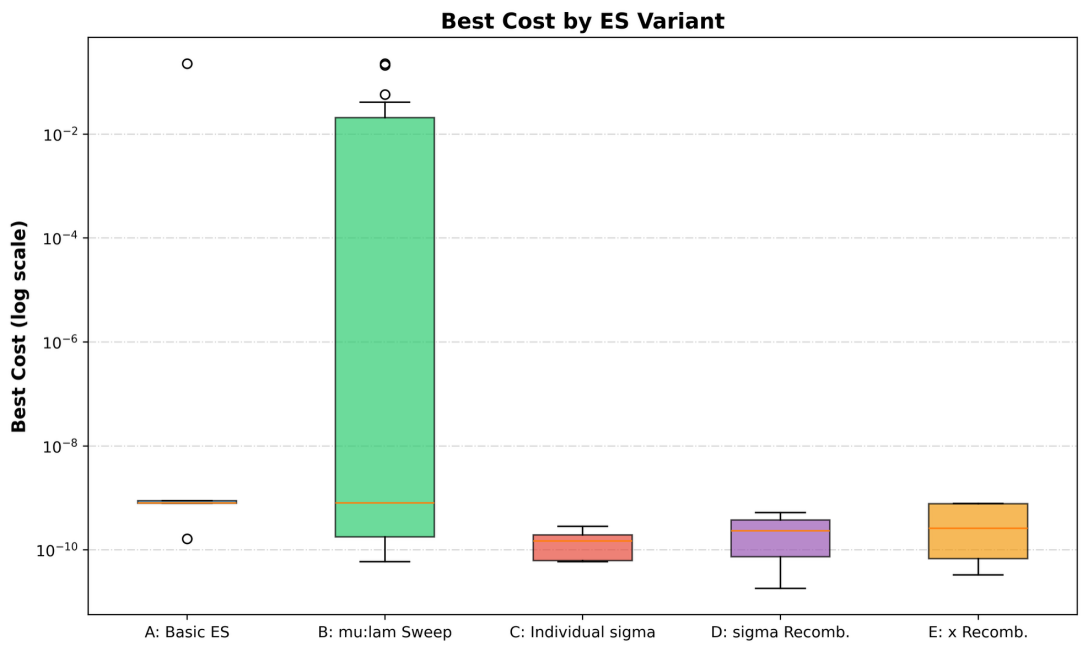

Figure 5 Best cost by ES variant

3.9. Convergence analysis

Figure 6 shows the convergence curves for all 35 runs grouped by experiment on a logarithmic scale. All variants exhibit rapid initial improvement in the first 50–100 generations, dropping from ~10^{1} to below 10^{-1}. In Experiment A, seed 3 plateaus near 10^{-1}, which indicats entrapment in a local minimum, while the other four seeds converge to ~10^{-9}. Experiment B shows a similar pattern: the smaller ratios (5:35, 15:100) have seeds that stagnate, whereas the 30:200 ratio converges reliably. Experiments C, D, and E (all using 30:200) display remarkably consistent convergence across all seeds, reaching 10^{-9} to 10^{-10} by generation 500. The staircase pattern

visible in the log scale curves reflects the periodic σ adaptation by the 1/5 rule every 10 generations.

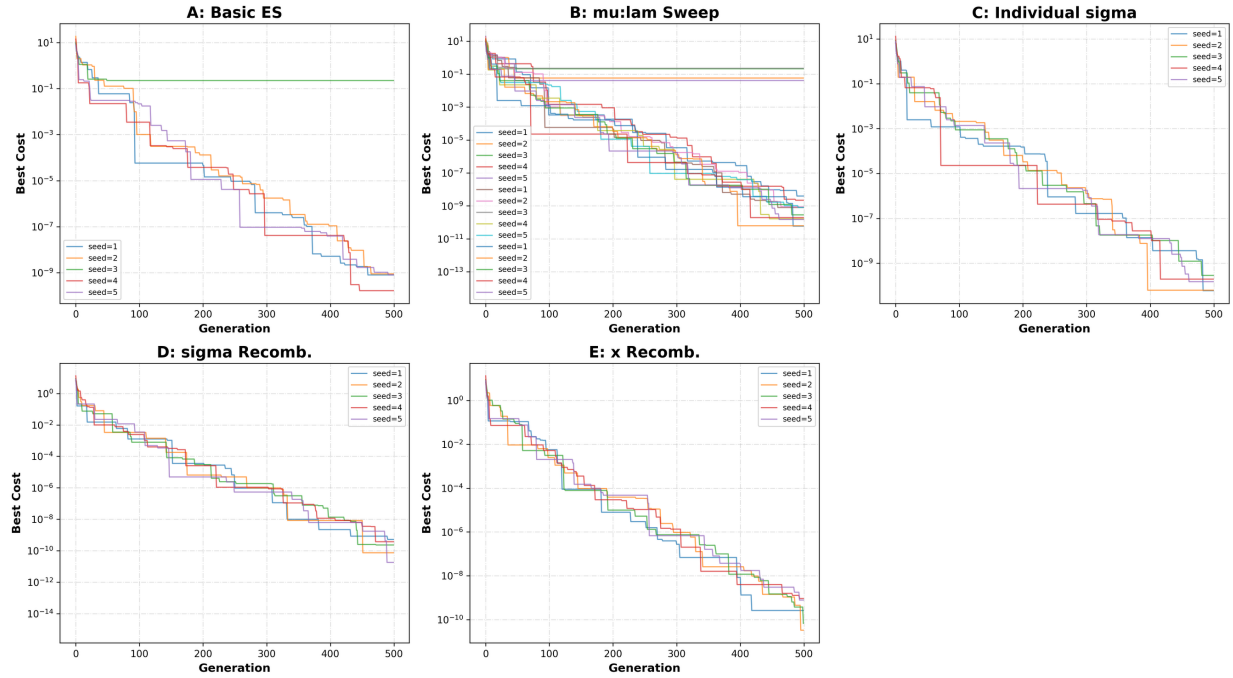

Figure 6 Convergence curves for all ES variants

3. 10. Best Solution across all 35 runs

The best solution across all 35 runs was found in Experiment D_sigma_recomb with seed 5, using μ:λ = 30:200. The solution is x* = (-0.00000002, 0.00000030) with f(x*) = 1.8122e-11, effectively the global optimum.

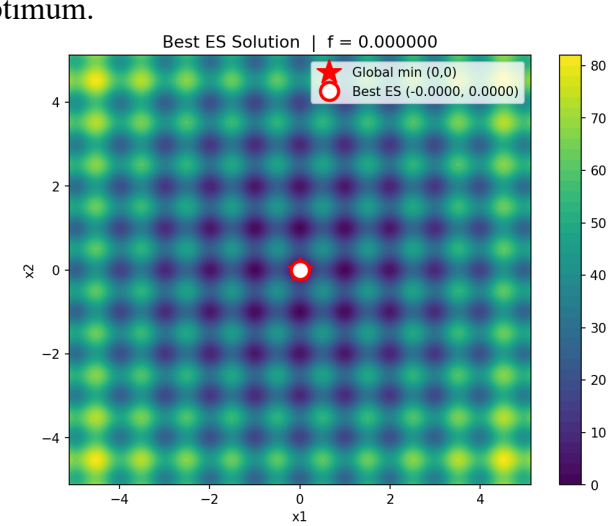

Figure 7 Best overall ES solution on the Rastrigin contour

3. 11. Runtime vs. cost tradeoff and summary of experiments

Figure 8 shows the runtime-versus-cost tradeoff. The basic ES and ratio sweep (A, B) complete fastest (mean 0.146s) but show greater cost variance. Experiments with recombination (D, E)

take longer (mean 0.751s) but achieve consistently near-zero cost. The log scale shows that the cost difference between trapped and successful runs spans roughly 8 orders of magnitude (10^{-2} vs. 10^{-10}), which underscoring the importance of sufficient population size.

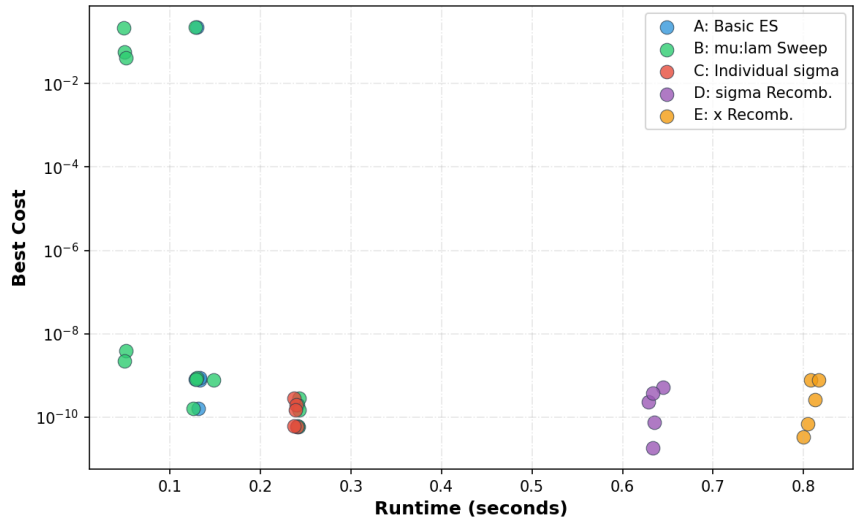

Figure 8 Runtime vs. best cost across all 35 runs

Table 1consolidates the performance of all five ES variants across the 35 runs. Experiments A and B, which use a single shared σ, exhibit mean costs on the order of 10^{-2}, driven by occasional seeds that become trapped in local minima (max cost ≈ 0.227). In sharp contrast, Experiments C, D, and E — all using the 30:200 ratio with individual σ — achieve mean costs on the order of 10^{-10}, with negligible standard deviations, confirming that every seed reliably converges to the global optimum. The best single run belongs to Experiment D (σ recombination), which reached $f^* = 1.81\times10^{-11}$. The runtime increases progressively from 0.131s (A) to 0.808s (E) as more sophisticated operators are added, but even the most expensive variant completes in under one second, making the computational overhead of recombination negligible relative to the improvement in robustness.

Table 8 Summary of all 35 ES runs

Experiment	***Num Runs***	***Mean Cost***	***Std***	***Min Cost***	***Max Cost***	***Mean Runtime (s)***
A: Basic ES	*5*	*0.045381891*	*0.101476991*	*1.64E-10*	*0.226909451*	*0.131*
B: μ:λ sweep	*15*	*0.035926013*	*0.076702153*	*5.92E-11*	*0.226909451*	*0.142*
C: Individual sigma	*5*	*1.50E-10*	*9.48E-11*	*5.92E-11*	*2.84E-10*	*0.239*
D: sigma Recomb	*5*	*2.45E-10*	*2.10E-10*	*1.81E-11*	*5.25E-10*	*0.635*
E: x Recomb	*5*	*3.84E-10*	*3.71E-10*	*3.31E-11*	*7.82E-10*	*0.808*

4. Conclusion

This study applied a (μ, λ)-Evolutionary Strategy to minimize the two-dimensional Rastrigin function, conducting 35 systematic runs across five progressively more sophisticated ES variants. The ES successfully found the global optimum f(0, 0) = 0 in the majority of runs, with the best result achieving f* = 1.8122e-11 at x* ≈ (0, 0). The key findings are: (1) The μ:λ ratio is the single most influential parameter. The 30:200 ratio consistently outperformed smaller ratios. (2) The 1/5 success rule provided effective step-size adaptation without manual σ tuning. (3) Individual σ per variable provided consistent improvement by allowing direction-specific adaptation. (4) Global intermediate recombination on σ produced the overall best solution. (5) Dual discrete recombination on x maintained reliable convergence. Overall, the results validate the (μ, λ)-ES with the 1/5 success rule as an effective method for continuous multimodal optimization.

5. Python code

The Python code is provided in the GoogleColab platform, as followg link:

Jupyter notebook Evolutionary Strategies.ipynb

References

1. Schönenberger, L., & Beyer, H. G. (2023, July). On a population sizing model for evolution strategies optimizing the highly multimodal rastrigin function. In *Proceedings of the Genetic and Evolutionary Computation Conference* (pp. 848-855).
2. Kauf, P. (n.d.). *Introduction to evolutionary strategies*. Kaggle. https://www.kaggle.com/code/pawelkauf/introduction-to-evolutionary-strategies
3. Ha, D. (2017, October 29). *A visual guide to evolution strategies*. 大トロ. https://blog.otoro.net/2017/10/29/visual-evolution-strategies/

Appendix

Table A1: All 35 ES runs detailed results

Run	***Experiment***	***Seed***	***μ:λ***	***Best x_1***	***Best x_2***	***Best Cost***	***RT (s)***
1	*A_basic*	*1*	*15:100*	*1.3987e-06*	*-1.4252e-06*	*7.9111e-10*	*0.149*
2	*A_basic*	*2*	*15:100*	*1.8180e-06*	*1.0706e-06*	*8.8313e-10*	*0.136*
3	*A_basic*	*3*	*15:100*	*2.5795e-02*	*-2.1922e-02*	*2.2691e-01*	*0.135*
4	*A_basic*	*4*	*15:100*	*-5.3746e-07*	*7.3345e-07*	*1.6404e-10*	*0.136*
5	*A_basic*	*5*	*15:100*	*-3.9903e-08*	*2.0127e-06*	*8.0399e-10*	*0.133*
6	*B_ratio*	*1*	*5:35*	*4.4304e-06*	*5.8950e-07*	*3.9630e-09*	*0.089*
7	*B_ratio*	*2*	*5:35*	*1.2653e-02*	*1.1538e-02*	*5.8149e-02*	*0.050*
8	*B_ratio*	*3*	*5:35*	*-9.0964e-03*	*-3.1488e-02*	*2.1248e-01*	*0.050*
9	*B_ratio*	*4*	*5:35*	*3.0909e-06*	*1.2465e-06*	*2.2037e-09*	*0.055*
10	*B_ratio*	*5*	*5:35*	*-1.4300e-02*	*-2.0271e-03*	*4.1357e-02*	*0.052*
11	*B_ratio*	*1*	*15:100*	*1.3987e-06*	*-1.4252e-06*	*7.9111e-10*	*0.129*
12	*B_ratio*	*2*	*15:100*	*1.8180e-06*	*1.0706e-06*	*8.8313e-10*	*0.135*
13	*B_ratio*	*3*	*15:100*	*2.5795e-02*	*-2.1922e-02*	*2.2691e-01*	*0.131*
14	*B_ratio*	*4*	*15:100*	*-5.3746e-07*	*7.3345e-07*	*1.6404e-10*	*0.135*
15	*B_ratio*	*5*	*15:100*	*-3.9903e-08*	*2.0127e-06*	*8.0399e-10*	*0.133*
16	*B_ratio*	*1*	*30:200*	*1.4021e-07*	*5.2807e-07*	*5.9224e-11*	*0.257*
17	*B_ratio*	*2*	*30:200*	*-5.0113e-07*	*-2.4881e-07*	*6.2105e-11*	*0.258*

18	*B_ratio*	*3*	*30:200*	*-1.0749e-06*	*5.2639e-07*	*2.8420e-10*	*0.253*
19	*B_ratio*	*4*	*30:200*	*-5.0849e-08*	*-9.8759e-07*	*1.9401e-10*	*0.252*
20	*B_ratio*	*5*	*30:200*	*8.6606e-07*	*4.6357e-08*	*1.4923e-10*	*0.255*
21	*C_indiv_sigma*	*1*	*30:200*	*1.4021e-07*	*5.2807e-07*	*5.9224e-11*	*0.271*
22	*C_indiv_sigma*	*2*	*30:200*	*-5.0113e-07*	*-2.4881e-07*	*6.2105e-11*	*0.245*
23	*C_indiv_sigma*	*3*	*30:200*	*-1.0749e-06*	*5.2639e-07*	*2.8420e-10*	*0.248*
24	*C_indiv_sigma*	*4*	*30:200*	*-5.0849e-08*	*-9.8759e-07*	*1.9401e-10*	*0.247*
25	*C_indiv_sigma*	*5*	*30:200*	*8.6606e-07*	*4.6357e-08*	*1.4923e-10*	*0.246*
26	*D_sigma_recomb*	*1*	*30:200*	*-8.7803e-07*	*-1.3688e-06*	*5.2467e-10*	*0.701*
27	*D_sigma_recomb*	*2*	*30:200*	*5.1446e-07*	*3.3120e-07*	*7.4273e-11*	*0.657*
28	*D_sigma_recomb*	*3*	*30:200*	*-8.1218e-07*	*7.1491e-07*	*2.3226e-10*	*0.658*
29	*D_sigma_recomb*	*4*	*30:200*	*-6.7504e-07*	*-1.1952e-06*	*3.7379e-10*	*0.663*
30	*D_sigma_recomb*	*5*	*30:200*	*-2.4737e-08*	*3.0123e-07*	*1.8122e-11*	*0.663*
31	*E_var_recomb*	*1*	*30:200*	*8.3139e-07*	*7.9141e-07*	*2.6139e-10*	*0.853*
32	*E_var_recomb*	*2*	*30:200*	*3.6610e-07*	*-1.8164e-07*	*3.3133e-11*	*0.824*
33	*E_var_recomb*	*3*	*30:200*	*-5.7540e-07*	*-1.1176e-07*	*6.8162e-11*	*0.822*
34	*E_var_recomb*	*4*	*30:200*	*1.9466e-06*	*3.5771e-07*	*7.7716e-10*	*0.841*
35	*E_var_recomb*	*5*	*30:200*	*-1.8230e-06*	*-7.8656e-07*	*7.8209e-10*	*0.827*